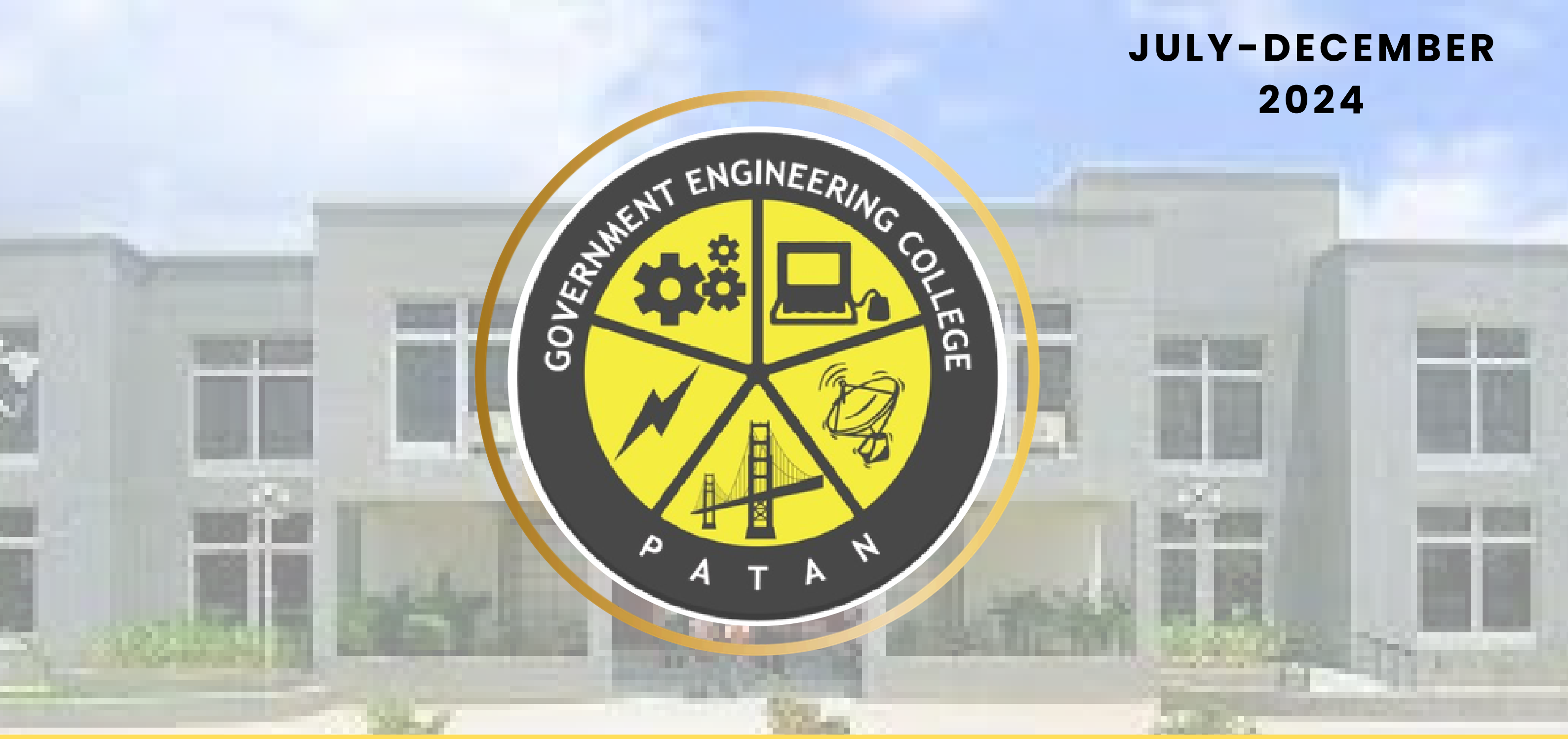
JULY-DECEMBER
2024



GOVERNMENT ENGINEERING COLLEGE PATAN

NEWSLETTER

INSTITUTE GLIMPSE

Government Engineering College, Patan (G.E.C. Patan), situated in the tranquil and scenic village of Katpur in Patan District, Gujarat, was established in 2004 with a commitment to delivering quality education in engineering and technology. Over the years, the institute has achieved remarkable growth and excellence in academics, innovation, and research. It is affiliated with the Gujarat Technological University (GTU) and functions under the administration of the Directorate of Technical Education (DTE), Government of Gujarat.

INSTITUTE VISION

“To prepare human resources with value-based competency for technical advancements and growth of society.”

INSTITUTE MISSION

- **To Deliver technical programs and services to cater the current needs of society and industry.**
- **Helping industries in solving challenges by means of providing best technical human resources.**
- **To contribute in sustainable growth of society.**

ELECTRONICS AND COMMUNICATION DEPARTMENT

NEWSLETTER

ABOUT

The Electronics and Communication Engineering (ECE) Department at Government Engineering College, Patan, established in 2004, offers a four-year undergraduate program with modern infrastructure, advanced laboratories, and experienced faculty. The department provides strong foundations in electronics, communication, networking, wireless technologies, and VLSI, preparing students for careers, research, and higher education.

DEPARTMENT VISION

"To prepare engineers with essential technical knowledge, value-based professional skills for technical upgradation and societal growth."

DEPARTMENT MISSION

- **To prepare competent engineers with the ability to solve industrial problems who can raise the living standards of the society.**
- **To prepare engineers with effective communication skills, professional ethics and leadership qualities.**

INDEX

Title	Page No.
Institute Vision Mission	1
Department Vision Mission	2
INDEX	3
HOD's Message About Department	4
PEOs- PSOs and POs	5 to 7
Faculty Details	8
Department Facilities	9 to 10
Department Activities	11 to 12
Committee Activities (NSS)	13 to 20

HOD’S MESSAGE

PROF. SANJAY D. JOSHI

It is my immense pleasure to present this Newsletter of the Electronics and Communication Engineering Department, Government Engineering College, Patan — a comprehensive documentation of our academic journey, industry exposure, and innovation across the years.

This edition reflects the relentless dedication of our faculty and the enthusiasm of our students. From impactful industry visits and expert lectures to project presentations and faculty development programmes, each activity represents our collective commitment to bridging academia and industry. I express my heartfelt gratitude to all stakeholders — our students, faculty, industry partners, and the college administration — who have continually contributed to making our department a centre of excellence.

~

Prof. Sanjay D. Joshi

HOD EC Department

Government Engineering College, Patan

ELECTRONICS AND COMMUNICATION ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

- Accumulate sound technical knowledge that enables to pursue higher education and research.
- Work in multidisciplinary areas with a strong focus on innovation and entrepreneurship.
- To inculcate effective communication skills, managerial skills, as well as professional ethics for successful professional career.

PROGRAM SPECIFIC OUTCOMES (PSOs)

- An ability to apply the conceptual knowledge of Microelectronics, Signal Processing, Microprocessors/Microcontrollers and Communication Systems to solve industrial problems.
- An ability to solve Electronics and communication Engineering problems using the latest hardware and software.

Program Outcomes (POs)

- **PO1: Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- **PO2: Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).
- **PO3: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).
- **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).
- **PO5: Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).
- PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

Program Outcomes (POs)

- **PO7: Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).
- **PO8: Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- **PO9: Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- **PO10: Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
- **PO11: Life-Long Learning:** Recognize the need for and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Faculty Members

PROF. SANJAY D. JOSHI
DESIGNATION: ASSOCIATE PROFESSOR
EMBEDED SYSTEM, IMAGE PROCESSING, MICROCONTROLLER AND MICOPROCESSOR

DR. HARESHKUMAR JUDAL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DIGITAL COMMUNICATION,WIRELESS COMMUNICATION AND DSP

PROF. MIHIRKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: DSP, COMMUNICATION SYSTEMS

DR. MAULIKKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: MICROWAVE, ANTENNA, ELECTRONICS

PROF. JAYESHKUMAR C. PRAJAPATI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: COMMUNICATION ENGG, MICROWAVE ENGG, OPTICAL FIBER COMMUNICATION

PROF.SHYAMKUMAR PANKHANIYA
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEM DESIGN, VLSI TECHNOLOGY

PROF. MEHULKUMAR PATEL
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: VLSI DESIGN

PROF. ROSHNIBEN CHAUDHARI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: ANTENNA, RF & MICROWAVE ENGINEERING, WIRELESS COMMUNICATION, BASIC ELECTRONICS

PROF. ANJU VASDEWANI
DESIGNATION: ASSISTANT PROFESSOR
AREA OF INTEREST: EMBEDDED SYSTEMS, DIGITAL CIRCUITS AND DESIGNING, MICROPROCESSOR, MICRO CONTROLLER & INTERFACING, PROGRAMMING USING C, BASIC ELECTRONICS

FACILITIES OF EC DEPARTMENT

L201 ANALOG & DIGITAL ELECTRONICS LABORATORY

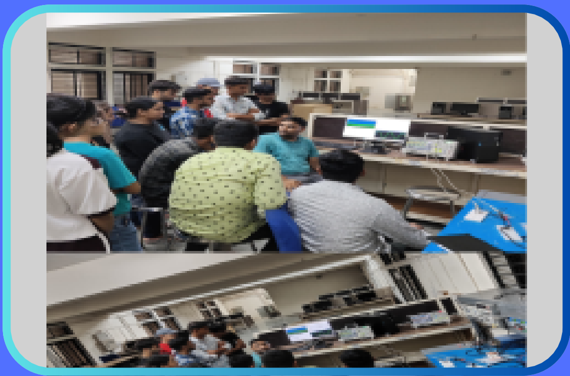
This laboratory is equipped with Analog and Digital Electronics trainers, viz.,amplifier, common emitter configuration, common collector configuration,oscillators, multivibrators, power supply, trainers using 555 IC, digital gates, flip flops, counters, and registers kits, etc. Following are few kits:

- Active band pass filter
- Automatic gain control kit
- Butterworth filter trainer
- CE configuration of transistor
- DC motor speed control using SCR

L202 SIMULATION LABORATORY

This laboratory is equipped with the latest computers with advanced simulation software, viz., MATLAB, Multisim, Visual Studio Code, Keil uVision, Atmel Studio and Xilinx etc. Laboratory is equipped with:

- DIGITAL OSCILLOSCOPE DSOX 2024A
- Workbench Electronics



L203 ELECTRONIC SYSTEM DESIGN LABORATORY

This laboratory consists of multiple electronics workbenches, having a digital storage oscilloscope, multi meter, function generator, and power supply, which can be helpful in testing and measurements of various electronics circuits or systems.

FACILITIES OF EC DEPARTMENT

L211 COMMUNICATION LABORATORY

This laboratory is equipped with Analog and Digital communication trainers, viz., AM, FM, and PM modulation demodulation, superheterodyne receiver, PWM, PPM, PAM, ASK, FSK, PSK kits, Spectrum Analyzer as well as RF and Microwave trainers for microwave communication along with active and passive components. Few trainers are as:

- Amplitude Modulation/Demodulation Trainer
- DSB/SSB AM Receiver
- Fiber Optics Demonstrator
- FM Transmitter Receiver Trainer
- Microwave Bench

L212 PROJECT LABORATORY

This laboratory is designed to provide students with hands-on experience in research and practical applications across various fields of Electronics and Communication Engineering.

L213 SIGNAL PROCESSING & VLSI LABORATORY

This laboratory consists of various trainers, viz., DSP, FPGA kits, and software required to perform operations on the trainers for Digital Signal Processing, Image Processing, Speech Signal Processing,
and VLSI experiments.

- DSP Trainer Kit
- Desktop Computer
  (ACER- AMD processor 12th generation)
- Programmable Function Generator, 1 Channel, 50Mhz (AFG3051C)

INDUSTRIAL VISIT

- The industrial visit to PCB Power was an enriching experience for the students, offering them firsthand exposure to the PCB manufacturing industry. The visit bridged the gap between theoretical knowledge and practical application, making it a highly informative and beneficial experience. The Electronics and Communication Engineering Department expresses gratitude to PCB Power for facilitating this valuable learning opportunity

Student Learnings

The visit provided students with valuable insights into the PCB manufacturing industry, including: Manufacturing processes: Students learned about the various stages involved in PCB fabrication and assembly. Quality control and testing: Students understood the importance of quality control in ensuring the reliability of PCBs. Industry trends and technologies: Students were exposed to the latest advancements in PCB manufacturing technology.

Expert Lecture

- On 14th October 2024, the Electronics and Communication Department of Government Engineering College Patan organized an expert lecture on "PCB Design," delivered by alumnus Mr. Soham Prajapati.
- The session aimed to bridge the gap between theoretical knowledge and practical application in PCB design, covering topics such as PCB types, design tools like Altium Designer and KiCad, design guidelines, manufacturing processes, and realworld case studies. Mr. Prajapati also shared his academic and professional journey, inspiring attendees and providing career tips. Organized by Prof. Shyam Pankhaniya, the lecture was attended by over 50 students and faculty members, who appreciated its depth and clarity. The event was a resounding success, leaving students motivated to explore opportunities in PCB design and modern electronics.

Tree Plantation Drive
Event Date : 10th July 2024 Event Venue : GECP College Campus

In response to the growing challenges posed by climate change and extreme weather events, the Hon'ble Prime Minister of India, Shri Narendra Modi, launched the "एक पेड़ माँके नाम" campaign, encouraging citizens to plant trees. On 10th July 2024, the NSS unit of Government Engineering College, Patan, organized a tree plantation drive on campus as part of this initiative. The event, attended by college principal Dr. B.J. Shah, faculty members, and NSS volunteers, involved the planting of 80 trees, including Aamla, Jambudo, and Gulmohar. The participants took an oath to nurture the trees, and plants were also distributed to local residents. The drive concluded with a briefing on environmental awareness by NSS Program Officer N.K. Dabhi Sir

Seedball Distribution Abhiyan

A seed ball is a low-cost and effective way of promoting fast plantation by encasing seeds in a mixture of soil and compost, protecting them from drying out or being eaten. On 10th July 2024, NSS volunteers from Government Engineering College Patan made 200 seed balls using seeds of plants like cabbage, okra, peas, and brinjal. These were distributed on 16th July 2024

Guru Purnima

Guru Purnima, a festival honoring spiritual and academic gurus, was celebrated by NSS volunteers at GEC Patan on 22nd July 2024.

The volunteers expressed their gratitude by felicitating respected faculty, administrative staff, and workers through a traditional Tilak ceremony. Divided into groups, they visited faculties, HODs, and the Principal, Dr. B.J. Shah, who offered his blessings for their bright future. The event highlighted the importance of gurus in our lives and reinforced the college's commitment to preserving cultural traditions and fostering a supportive academic community

Fire Safety Awareness Programme

Fire safety measures are crucial for saving lives, protecting property, and ensuring legal compliance through early warnings like alarms and suppression systems that help control fires. To raise awareness of these measures, NSS GEC Patan organized a fire safety awareness program on 26th July 2024, led by Fire Officer Snehal Modi from Patan Station. The session, guided by a PowerPoint presentation, covered essential fire prevention techniques, the use of fire extinguishers, and evacuation procedures.

Nasha Mukti Abhiyan

Government Engineering College Patan in collaboration with District Officer Nilesh Desai, dedicated to ensuring children's safety and well-being had organized a Nasha Mukti Bharat Abhiyan event. This initiative aims to educate and aware children about the dangers of substance abuse and addiction, promoting a healthier and safer lifestyle.

The event was organized at Electrical Seminar Hall, GEC Patan. The event commenced at 11 a.m. First, we welcomed our all guest with flower bouquet. The event started by deep pragtya, with a soulful rendition of NSS song. Afterwards, our District Officer Nilesh Desai and our professor gave the speech on Nasha Mukti Bharat Abhiyan. Then we pledged to uphold the cause of nasha mukti, dedicating ourselves to creating a society free from the grip of addiction. The event concluded by the resonation of National Anthem.

The campaign reflects the government's commitment to addressing the growing issue of substance abuse, which poses significant social, economic, and health challenges to the country. Through interactive sessions, expert talks, and engaging activities, we strive to empower children with knowledge and skills to make informed choices, fostering a secure and nurturing environment for their growth and development. Together, we can create a brighter, addiction-free future for our young generation.

Anti-Ragging Awareness Event

On 13th August 2024, Government Engineering College, Patan, observed AntiRagging Day, organized by the NSS unit to foster a harmonious, ragging-free environment for students, particularly the new entrants. The event, held in the presence of the principal, faculty, and newly admitted students, began at 11:00 am with the melodious NSS song, followed by a warm welcome to the chief guest. The chief guest addressed the gathering, discussing the negative impact of ragging on student life and offering preventive measures to ensure a safe and supportive campus environment.

The college principal reinforced the institution's commitment to preventing ragging and assured continuous awareness efforts to uphold a ragging-free campus. Throughout the event, students creatively conveyed the anti-ragging message through performances, songs, and speeches, encouraging unity and respect among peers. The program concluded with an oath of "Anti-Ragging," affirming the collective dedication of students and staff to fostering a safe, inclusive space for all. The event was a resounding success, leaving a lasting impact on the participants.

Swachhta Jumbesh 4.0 (Phase-2)

The Swachhata Jumbesh 4.0 campaign, conducted by the Indian Government from October 1 to 30, focused on cleanliness and minimizing pendency across higher education institutions. NSS GEC Patan actively participated in this initiative with a month-long cleanliness drive, dividing efforts into four weeks.

In Week 1, volunteers cleaned the main campus, focusing on pathways, administrative blocks, and nonbiodegradable waste while raising awareness among students and faculty. Week 2 emphasized cleaning classrooms, laboratories, and department surroundings, ensuring hygienic learning spaces. Week 3 targeted the New Amenity Block, canteen, seminar halls, and outdoor areas, promoting environmental protection by maintaining greenery and proper waste disposal. Finally, Week 4 revisited the campus to ensure sustained cleanliness, motivating staff and students to uphold hygiene practices. The campaign successfully fostered a lasting culture of cleanliness and environmental responsibility within the campus community.

Vikas Saptah Ujavani

Bharat Vikas Saptah was celebrated from 7th and 15th October 2024 with the competitions and various activities of developmental achievements and goals. The events aimed to raise awareness about "Viksit Bharat 2047" and promote the vision of "Ek Bharat Shreshtha Bharat." Activities included an elocution competition on the theme "Ek Bharat Shreshtha Bharat" and an oath ceremony (Bharat Vikas Shapath Vidhi) for new students. The events witnessed active participation from 197 students in two days, conducted offline in the Electrical Seminar hall, fostering engagement and enthusiasm among participants

National Integration Camp

The National Integration Camp (NIC) 2024, hosted by Guru Jambeshwar University of Science & Technology in Hisar from October 11-17, brought together over 200 participants from 20+ universities across 16 states, fostering cultural understanding and national unity. The camp, organized by the National Service Scheme (NSS), featured a delegation from Gujarat Technological University, led by Program Officer Nimesh Prajapati, along with eight volunteers. Throughout the week, participants engaged in technical sessions, cultural exchanges, workshops, and excursions, celebrating India's rich heritage and diversity. Activities included discussions on leadership, digital literacy, and legal literacy, as well as creative competitions, sports events, and a memorable visit to Kurukshetra to explore its historical and spiritual significance.

The camp concluded with impactful events like a rally, cultural performances, and community service, instilling a sense of social responsibility and patriotism. The valedictory ceremony highlighted the achievements of participants and emphasized the importance of national integration

GEC PATAN YOUTH PARLIAMENT

On October 18, 2024, Government Engineering College, Patan, hosted a Youth Parliament event as part of the 4th Edition of the National Youth Parliament Scheme (NYPS), an initiative by the Government of India. Organized under the leadership of Chief Patrons Dr. Bharat J. Shah and Dr. Anand B. Dhruv, and guided by Dr. A.K. Chaudhary and Dr. Harshad Bhutadia, the event provided students with an immersive experience in parliamentary procedures. It aimed to instill leadership values, promote active citizenship, and enhance public speaking and debating skills. The opening ceremony featured keynote addresses by Dr. A. K. Chaudhary, who emphasized the role of youth in nationbuilding, and Dr. H. S. Bhutadia, who encouraged students to voice their opinions confidently

The event adhered to official parliamentary protocols, with 41 students assuming roles such as Speaker, Prime Minister, Opposition Leader, and Members of Parliament. Discussions covered topics like health concerns, climate change, national security, digital economy, and gender equality.

National Integration Camp

This year National Integration Camp 2024, was organized by Government of India Ministry Of Youth Affairs & Sports, NSS Regional Directorate Ahmedabad and Hemchandracharya North Gujarat University, Patan from 8th to 14th November, 2024. A total of 10 Volunteers Selected from Gujarat Technological UniversityAhmedabad. There were 2 Volunteers from GEC-PATAN(Dev Sharma and Krisha Panchal), 2 Volunteers from GEC-GODHRA, 2 Volunteers from APC-ANAND, 2 Volunteers from GPC-DAHOD, 1 Volunteer from SPEC ANAND, 1 Volunteer from VGEC-AHMEDABAD. In this Camp a total of 210 Volunteers were present from 10 different states and 1 Union Territory of India including Assam, Andhra Pradesh, Chhattisgarh, Haryana, Orissa, Maharashtra, Rajasthan, Madhya Pradesh, West Bengal, Gujarat and Dadra Nagar Haveli and Diu Daman. The day starts at 6:30 in the morning with a one-hour yoga session, followed by a workshop session of 2 hours on topics such as Sand Art, Photography, Garba, Self-Defense. After that an academic session was held by different distinguished speakers on various topics. After lunch, volunteers divided in groups to share their views on the topics as per schedule.

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HOD, Electronics and Communication Department

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